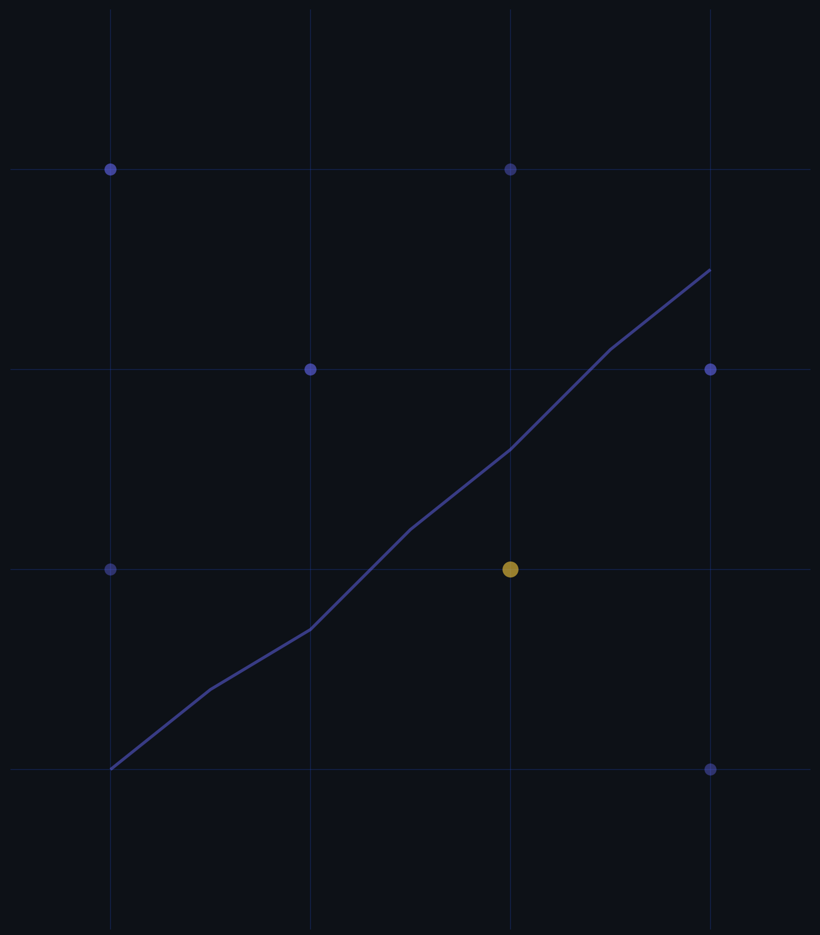
Global CO_2 Emissions: Who Emits, How Much, and What’s Changing

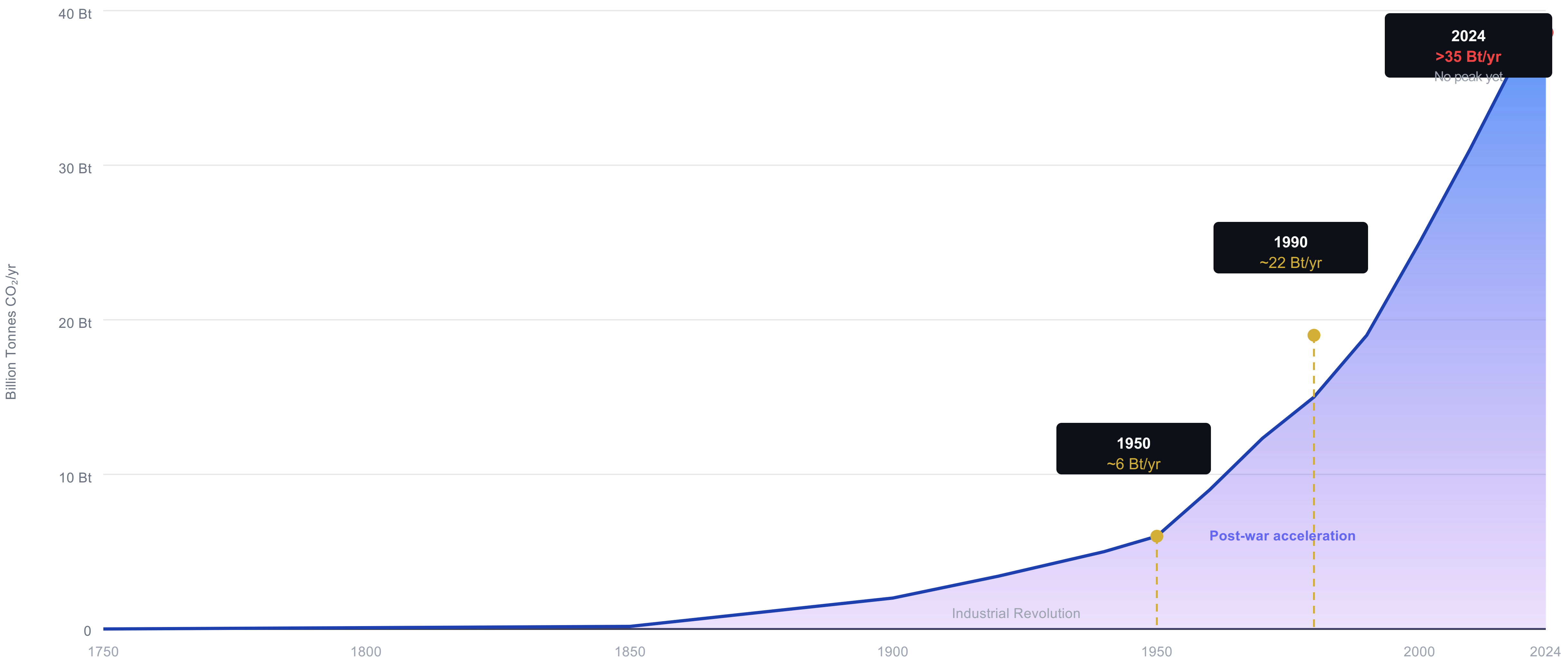
Fossil Fuels & Industry · 1750–2024 · Global Carbon Budget 2025 via Our World i



Data: Global Carbon Budget 2025, Our World in Data | Authors: Hannah Ritchie & Max Roser

Global CO_2 from fossil fuels exceeds 35 Bt/yr — and has not yet peaked

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Fossil fuel emissions dominate; land-use change has slightly declined in recent years.

Asia now produces ~50% of global CO_2 — a complete reversal from 1900

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

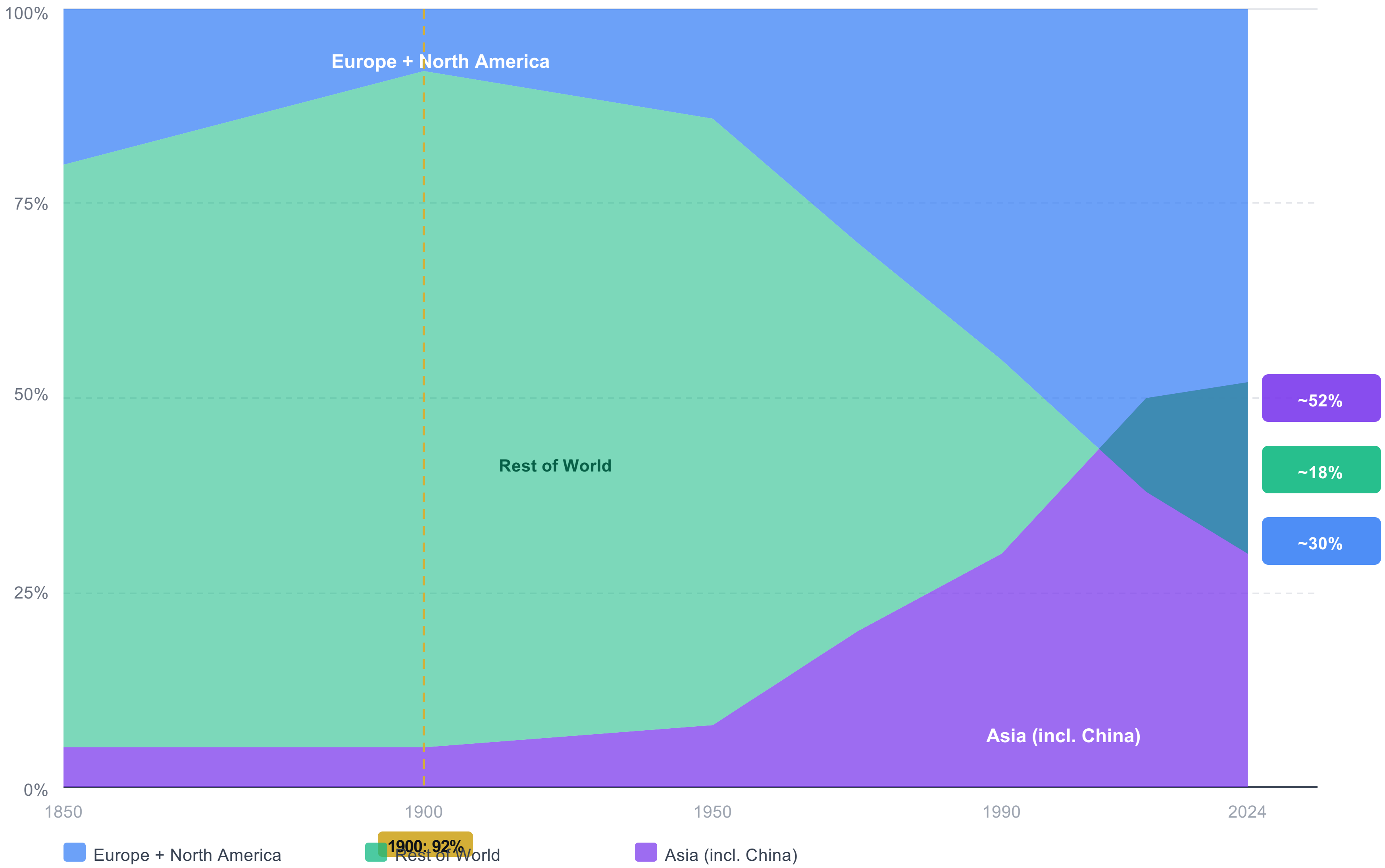
Historical Share Shift

1900
>90%
Europe + US share of global CO_2
Asia share: <5%

1950
>85%
Europe + US still dominant
Post-war reconstruction driving growth

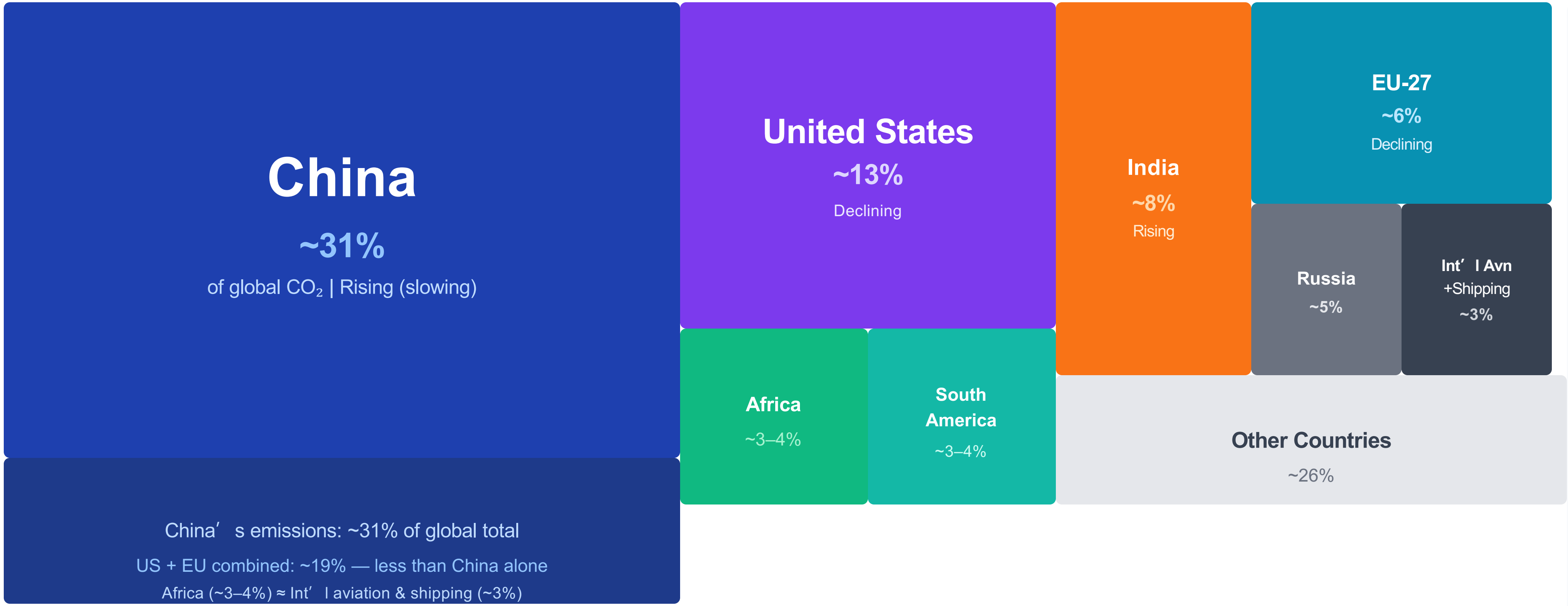
Today (2024)
<33%
Europe + US share
Asia: ~50% | China alone: ~31%

Strategic Implication
Climate responsibility debates increasingly involve historical emissions, not just current shares.



China alone accounts for ~31% of global CO_2 — more than the US and EU combined

China Now Emits More Than One-Quarter of Global CO_2



2024 Trends

China ~31%
Rising but slowing

US ~13%
Declining trend

India ~8%
Rapidly rising

EU-27 ~6%
Declining trend

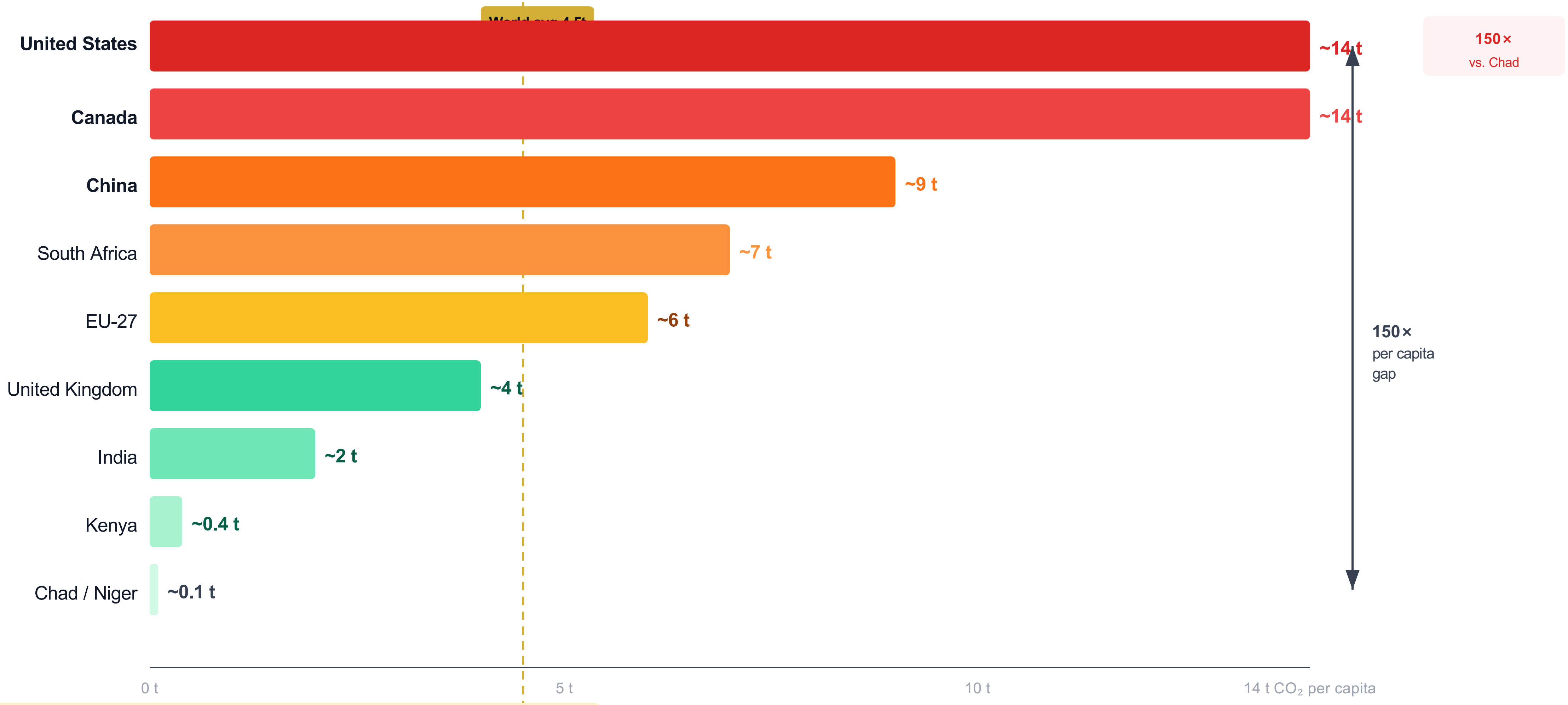
Russia ~5%
Roughly stable

Africa ~3–4%
Slowly rising

Source: Global Carbon Budget 2025 via Our World in Data (Ritchie & Roser) | Approx. 2024 shares

A 150× gap separates the richest and poorest nations — the average American emits in <2 days what a person in Chad emits in a year

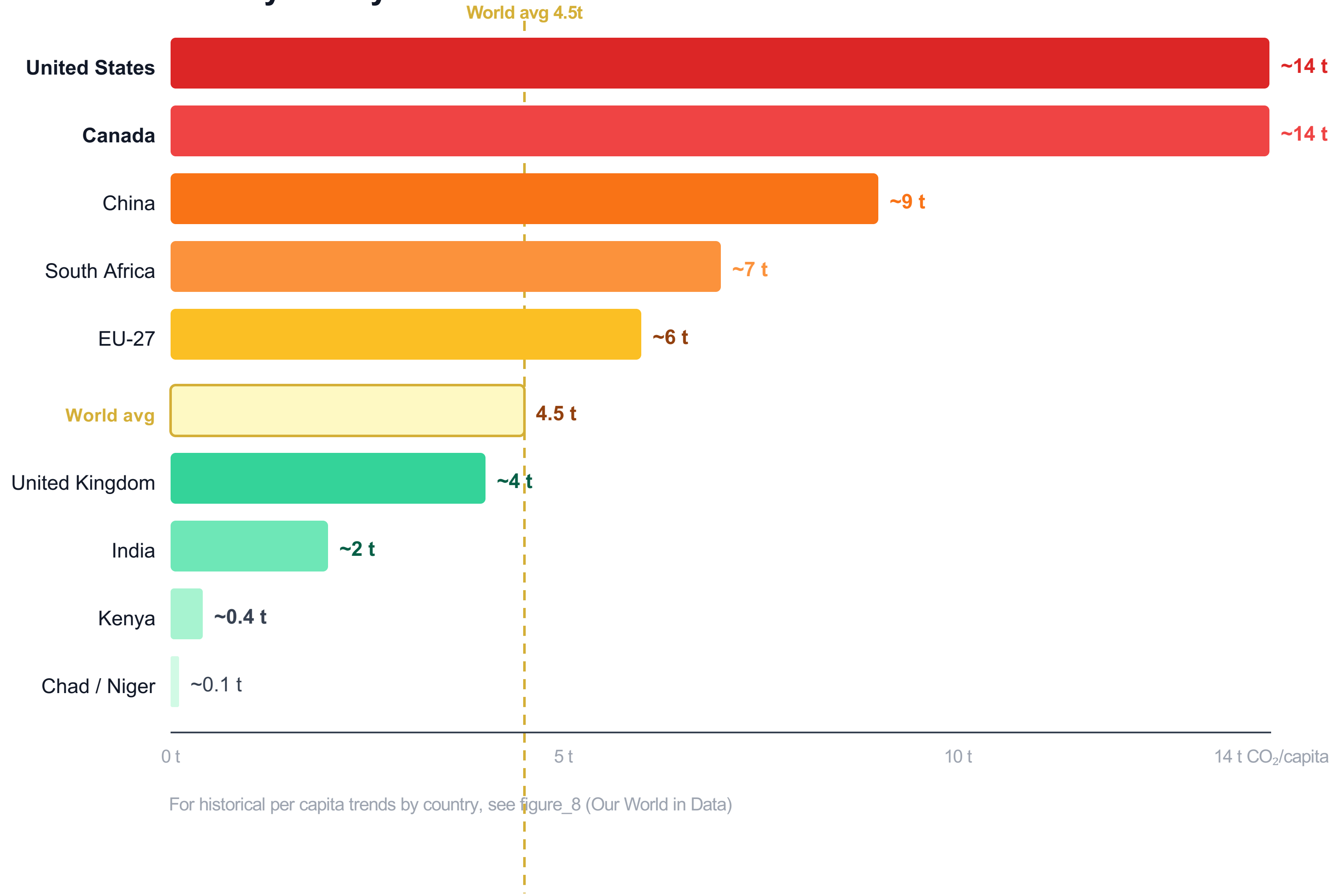
Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



Key inequality:
The average American emits in under 2 days what a person in Mali or Niger emits in an entire year.

China is the world' s #1 total emitter — yet per capita it emits only ~9 t, less than two-thirds of the US

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India



For historical per capita trends by country, see figure_8 (Our World in Data)

Key Insights

US & Canada: ~14 t

~3× the global average of 4.5 t/capita. Among the highest of any major economy globally.

China: ~9 t

Largest total emitter (~31% global) but per capita still below the US. Below US by ~35%.

India: ~2 t

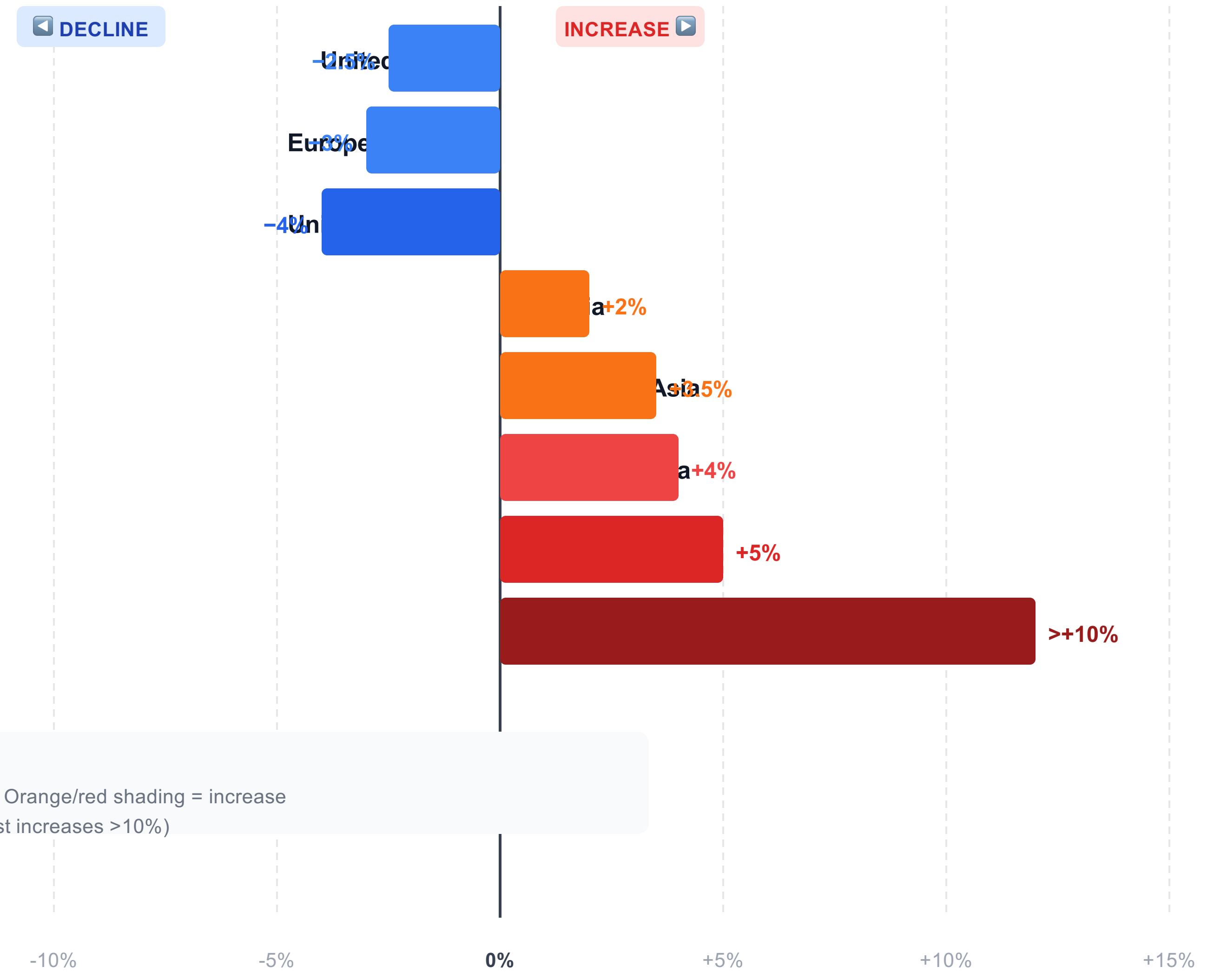
3rd largest total emitter (~8%) yet 7× below the US per capita. Below world avg of 4.5 t.

EU-27: ~6 t

Declining. ~1.3× global average. Lower than US despite similar GDP.

2024 shows a mixed picture: Western declines offset by rising emissions in Asia, the Middle East, and Africa

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew



Map reference (figure_5):
Blue shading = emissions decline (US, EU, UK); Orange/red shading = increase (Russia, SE Asia, Africa, Central Asia with largest increases >10%)

Net Assessment

Developed nation declines partially offset by rises in Asia, Middle East & Africa.

Global total: still rising
No peak reached in 2024

DECLINING:

- US, EU, UK, Japan
- Most W. Europe

INCREASING:

- Russia, SE Asia
- Parts of Africa
- Central Asia >10%

Similar prosperity, very different emissions — energy mix is the decisive variable

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Why the Gap?

The Mechanism

Countries with similar GDP/capita can have 3–4× different per capita CO_2 based on whether their electricity comes from fossil fuels or not.

Low Per Capita Emitters among Wealthy Nations

France, UK, Portugal

→ High nuclear + renewable share in electricity
UK ~4 t/cap | France ~4 t/cap (illustrative)
Portugal: below EU-27 average (~6 t)

Higher Per Capita Emitters among Wealthy Nations

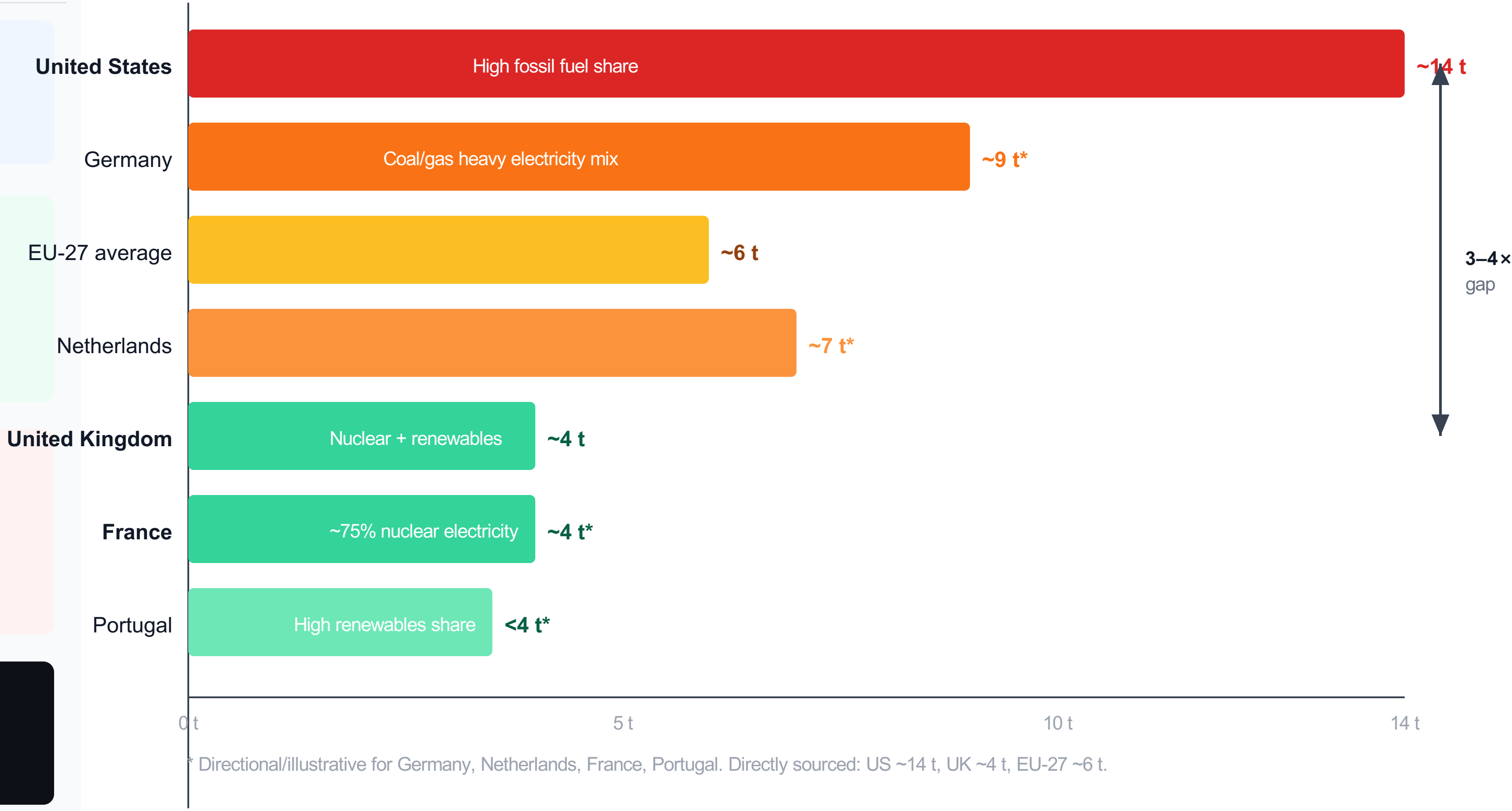
United States, Germany, Netherlands

→ Higher fossil fuel share in electricity & heating
US ~14 t/cap | Germany: above EU-27 avg
Netherlands: above EU-27 average

Policy Implication

Energy policy choices — not just economic development — can decouple prosperity from carbon emissions.

Per Capita CO_2 Emissions — Comparable Wealthy Nations (2024, approx.)



* Directional/illustrative for Germany, Netherlands, France, Portugal. Directly sourced: US ~14 t, UK ~4 t, EU-27 ~6 t.

Source: Global Carbon Budget 2025 via Our World in Data | Sourced values: US ~14 t, UK ~4 t, EU-27 ~6 t; others directional/illustrative

Key Takeaways: Responsibility Is Shared but Unequal

01 No peak yet

Global CO_2 from fossil fuels & industry exceeds 35 Bt/yr.
Growth has slowed but emissions have not peaked —
up from 6 Bt in 1950 and 22 Bt in 1990.

→ The long-run trajectory remains upward.

02 China dominates total, not per capita

China emits ~31% of global CO_2 (largest single share).
Yet per capita: ~9 t — below the US (~14 t) by ~35%.
India emits ~8% total but only ~2 t per capita.

→ Total vs. per capita tell very different stories.

03 Historical responsibility: Europe & the US

Europe and the US emitted >90% of global CO_2 before 1900.
Still >85% in 1950. Today: under one-third combined.
Cumulative historical stocks drive most current warming.

→ Historical emissions shape climate justice arguments.

04 150× per capita gap persists

Chad & Niger: ~0.1 t/person/yr. US & Canada: ~14 t/person.
The average American emits in <2 days what a person in
Mali emits in an entire year.

→ Per capita inequality is a central climate justice issue.

05 Energy policy can decouple prosperity from emissions

France and the UK achieve similar living standards to the US
but emit 3–4× less per capita — due to nuclear & renewables.
Technology and policy choices, not just wealth, determine
a country' s carbon footprint.

→ The path to net zero requires both ambition and the right energy mix.

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